Last Time

For a discrete random variable X,

$$E[X] = \sum_{x \in \mathcal{R}_X} x p_X(x).$$

For a continuous random variable X,

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

The expected value of X represents a probability-weighted average of the possible values of X, and can be interpreted as the long-run average over repeated experiments.

Mixed Type : discrete component + continuous component

probability - weighted average

Question: What if we are interested not in X, but in a function of X?

For example, if X is voltage, perhaps we need X^2 to calculate power.

PMF for discrete RVs
PDF for continuous RVs
CDF for all RVs

Expected Value of a Function of a Random Variable

Suppose

$$Y = g(X).$$

We could first determine the distribution of Y and then calculate $E[Y]$. However, this is often unnecessary.

Results

If X is discrete,

$$E[g(X)] = \sum_{x \in \mathcal{R}_X} g(x) p_X(x)$$

and if X is continuous,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx.$$

Thus, $E[g(X)]$ can be calculated directly from the distribution of X, without first finding the distribution of $Y = g(X)$.

Example. [Discrete Random Variable] A fair six-sided die is rolled. Let X be the number obtained. Suppose

$$Y = X^2.$$

Find $E[Y]$.

$$E[Y] = E[X^2]$$

$$= \sum x^2 p_X(x)$$

$$= \left(1^2 \cdot \frac{1}{6} + 2^2 \cdot \frac{1}{6} + \cdots + 6^2 \frac{1}{6}\right)$$

$$= 91/6$$

Note:

- we didn't need to first find the PMF of Y.
- $E[X^2] \neq (E[X])^2$ (check page 6.)

Example. [Continuous Random Variable] Let X be uniformly distributed on $[0, 2]$, with PDF

$$f_X(x) = \begin{cases} \frac{1}{2}, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find $E[X^2]$.

(b) Find the CDF and PDF of the random variable $Y = X^2$.

(c) Using the PDF of Y obtained in part (b), compute $E[Y]$ and verify your answer to part (a).

(a) $E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx = \int_0^2 x^2 \cdot \frac{1}{2}\,dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_0^2 = \frac{4}{3}$

(b) For $0 \le X \le 2$, we have $0 \le Y \le 4$.

For $0 \le y \le 4$, we have the CDF of Y:

$$F_Y(y) = P(Y \le y)$$

$\{Y \le y\}$

$$= P(X^2 \le y)$$

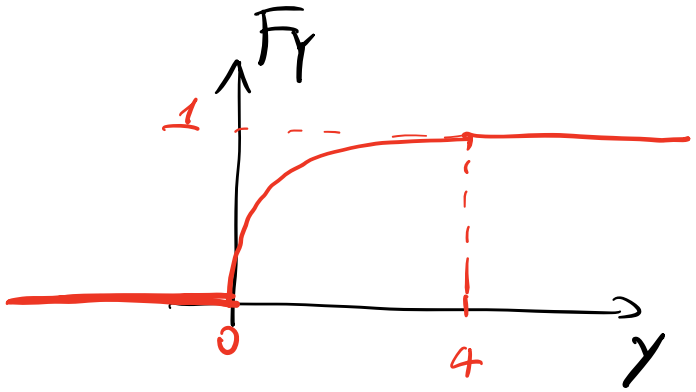
$\Leftrightarrow \{X^2 \le y\}$

$$= P(-\sqrt{y} \le X \le \sqrt{y})$$

$\Leftrightarrow \{-\sqrt{y} \le X \le \sqrt{y}\}$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x)\,dx$$

$$= \int_0^{\sqrt{y}} \frac{1}{2}\,dx = \frac{\sqrt{y}}{2}$$



Therefore, the CDF of Y is

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ \sqrt{y}/2, & 0 \le y \le 4 \\ 1, & y > 4 \end{cases}$$

Differentiating the CDF gives the PDF:

$$f_Y(y) = \frac{d}{dy}\left(F_Y(y)\right)$$

$$\therefore \quad f_Y(y) = \begin{cases} \frac{1}{4\sqrt{y}}, & 0 < y < 4 \\ 0, & \text{otherwise} \end{cases}$$

(c). $E[Y] \overset{\text{defn.}}{=} \int_{-\infty}^{\infty} y\, f_Y(y)\, dy$

$$= \int_0^4 y \cdot \frac{1}{4\sqrt{y}}\, dy$$

$$= \frac{1}{4}\int_0^4 \frac{1}{\sqrt{y}}\, dy$$

$$= \frac{4}{3}$$

which verifies the result obtained in part (a).

Show $E[b] = b$.

$E[b] = \sum b\, p_X(x) = b \sum p_X(x) = b \cdot 1 = b$

Linearity of Expectation

For constants a and b,

$$E[aX + b] = aE[X] + b.$$

More generally,

$$E[aX + bY] = aE[X] + bE[Y].$$

Variance

mean

Expected value describes the "**center**" of a random variable.

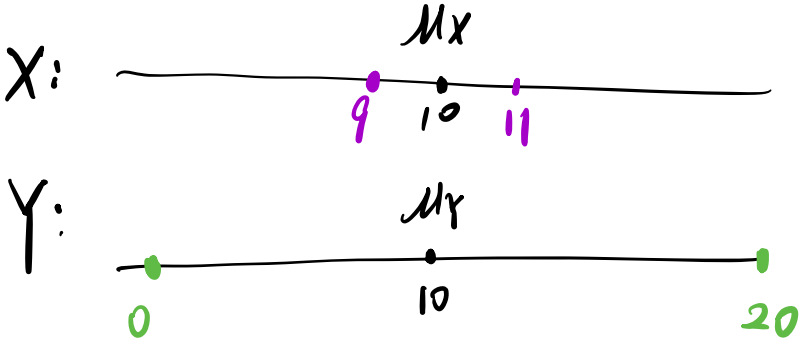
However, two random variables can have the same expected value while having very different distributions.

For example, suppose X takes the values 9 and 11 with equal probability, while Y takes the values 0 and 20 with equal probability.
Then

$0 \cdot \frac{1}{2} + 20 \frac{1}{2} = 10$

$$E[X] = E[Y] = 10.$$

However, Y is clearly much more spread out than X.



Question: How can we measure how spread out a random variable is around its mean?

The deviation of X from its mean:

$$X - \mu_X$$

Then $E[X - \mu_X] = E[X] - \mu_X$

$= E[X] - E[X]$

$= 0$

∴ The expected deviation $E[X - \mu_X]$ cannot measure the dispersion of X around μ_X.

Variance

The **variance** of random variable X is

$$\text{Var}[X] = E[(X - \mu_X)^2].$$

The variance measures the average squared deviation of X from its mean.

Computing Variance

For a discrete random variable,

$$\text{Var}[X] = \sum_{x \in \mathcal{R}_X} (x - \mu_X)^2 p_X(x).$$

For a continuous random variable,

$$\text{Var}[X] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x)\, dx.$$

There is also a useful alternative formula:

$$\text{Var}[X] = E[(X - \mu_X)^2]$$

function of X: $g(X) = (X - \mu_X)^2$

$$= E[X^2 - 2X\mu_X + \mu_X^2] \quad = X^2 - 2X\mu_X + \mu_X^2$$

$$= E[X^2] - 2\mu_X E[X] + \mu_X^2$$

$$= E[X^2] - 2E[X] \cdot E[X] + (E[X])^2$$

$$= E[X^2] - (E[X])^2$$

$$\therefore \quad \text{Var}[X] = E[X^2] - (E[X])^2$$

Example. Suppose X is the voltage measured at a sensor, with

$$P(X = 4 \text{ V}) = 0.5, \qquad P(X = 6 \text{ V}) = 0.5.$$

Then $E[X] = \sum x p_X(x) = 4 \cdot \frac{1}{2} + 6 \cdot \frac{1}{2} = 5 \text{ V}$

$E[X^2] = \sum x^2 p_X(x) = 4^2 \frac{1}{2} + 6^2 \frac{1}{2} = 26 \text{ V}^2$

and $\text{Var}[X] = E[X^2] - (E[X])^2 = 26 - 25 = 1 \text{ V}^2$

Variance measures spread, but it has squared units.

Standard Deviation

Standard Deviation

The **standard deviation** of X is defined as

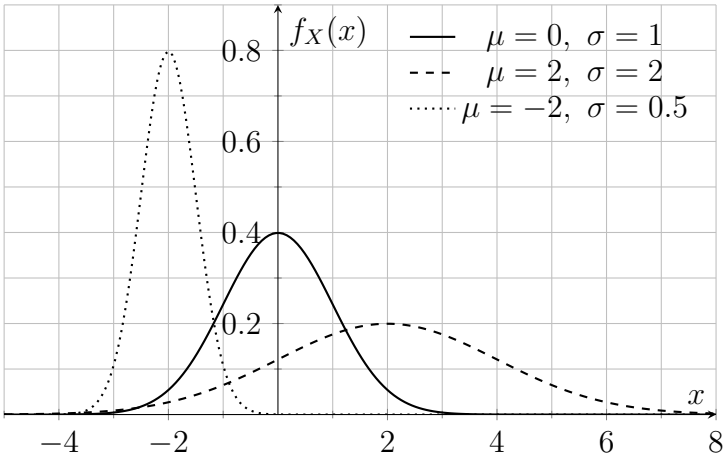
$$\sigma_X = \sqrt{\text{Var}[X]} = \sqrt{E[(X-\mu_x)^2]}.$$

The standard deviation has the same units as X.

μ_X	:	center of the distribution,
σ_X	:	spread of the distribution.

Recall: Gaussian Random Variable: $X \sim N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2},$$



If $X \sim N(\mu, \sigma^2)$,

$$\mu_X = \mu \qquad \sigma_X = \sigma$$

Expected value — Standard deviation

Properties of Variance

Suppose

$$Y = aX + b.$$

Then

$$E[Y] = aE[X] + b.$$

For the variance,

$$\operatorname{Var}[aX + b] = a^2 \operatorname{Var}[X].$$

Consequently,

$$\sigma_{aX+b} = |a|\sigma_X.$$

Moments of a Random Variable

The nth **moment** (or nth raw moment) of X is

$$E[X^n].$$

For example,

$$\begin{aligned} E[X] &: \text{ first moment,} \\ E[X^2] &: \text{ second moment.} \end{aligned}$$

The nth **central moment** is

$$E[(X - \mu_X)^n].$$

Therefore,

$$\operatorname{Var}[X] = E[(X - \mu_X)^2]$$

is the **second central moment** of X.

The terminology comes from mechanics, where moments describe how mass is distributed. Here, probability plays a role analogous to mass.